

SIMATS ENGINEERING

ASSESSMENT TOOL 4

Course Name: Artificial Intelligence | Course Code: CSA17

Course Outcome: CO3 — Knowledge Representation & Reasoning

Annexure A — Resolution Algorithm Implementation

ANSWER SCRIPT

Submitted by

S.KARTHIKEYAN

Register No: 192419048

Time Duration: 1 Hour | Total Marks: 30

Declaration:

I, S.KARTHIKEYAN (Reg No: 192419048), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Question 1 — Rain and Wet Ground

Knowledge Base

- If it rains, then the ground becomes wet.
- It is raining.

Goal: Prove that the ground is wet, using the Resolution Algorithm.

Step 1: Propositional Logic Representation

Let the atomic propositions be:

- R = It is raining
- W = The ground is wet

The knowledge base is written as:

Premise 1: $R \rightarrow W$

Premise 2: R

Goal: W

Step 2: Convert to Conjunctive Normal Form (CNF)

An implication $P \rightarrow Q$ is logically equivalent to $\neg P \vee Q$. Applying this to Premise 1:

$R \rightarrow W \equiv \neg R \vee W$

Premise 2 (R) is already a single literal, so it is already in CNF.

CNF clause set:

C1: $\neg R \vee W$

C2: R

Step 3: Negate the Goal

To prove W by contradiction (refutation), we negate the goal and add it to the clause set:

C3 ($\neg$ Goal): $\neg W$

Step 4: Apply the Resolution Algorithm

We resolve pairs of clauses that contain complementary literals until the empty clause is derived.

Resolve C2 & C1: R and $\neg R \vee W \rightarrow W$ (call this C4; R and $\neg R$ cancel)

Resolve C4 & C3: W and $\neg W \rightarrow \square$ (Empty Clause)

Step 5: Result

Since the resolution process derives the Empty Clause ($\square$), the negated goal $\neg W$ is contradictory with the knowledge base. Therefore the original goal is PROVED:

$\therefore$ The ground is wet. (Goal successfully proved by resolution refutation.)

Question 2 — Student Assignment Submission

Knowledge Base

- If a student submits the assignment, then the student receives internal marks.
- Rahul submitted the assignment.

Goal: Prove that Rahul receives internal marks, using the Resolution Algorithm.

Step 1: Propositional Logic Representation

Let:

- S = Rahul submits the assignment
- M = Rahul receives internal marks

Premise 1: $S \rightarrow M$

Premise 2: S

Goal: M

Step 2: Convert to CNF

$S \rightarrow M \equiv \neg S \vee M$

CNF clause set:

C1: $\neg S \vee M$

C2: S

Step 3: Negate the Goal

C3 ($\neg$ Goal): $\neg M$

Step 4: Apply the Resolution Algorithm

Resolve C2 & C1: S and $\neg S \vee M \rightarrow M$ (call this C4)

Resolve C4 & C3: M and $\neg M \rightarrow \square$ (Empty Clause)

Step 5: Conclusion

The derivation of the Empty Clause confirms that $\neg M$ contradicts the knowledge base. Hence the goal is PROVED:

$\therefore$ Rahul receives internal marks. (Goal successfully proved.)

Question 3 — Library Membership

Knowledge Base

- If a person is a library member, then the person can borrow books.
- Priya is a library member.

Goal: Prove that Priya can borrow books, using the Resolution Algorithm.

Step 1: Propositional Logic Representation

- L = Priya is a library member
- B = Priya can borrow books

Premise 1: $L \rightarrow B$

Premise 2: L

Goal: B

Step 2: Convert Each Statement to CNF

$L \rightarrow B \equiv \neg L \vee B$

CNF clause set:

C1: $\neg L \vee B$

C2: L

Step 3: Negate the Goal

C3 ($\neg$ Goal): $\neg B$

Step 4: Resolution Until No New Clauses Can Be Generated

Resolve C2 & C1: L and $\neg L \vee B \rightarrow B$ (call this C4)

Resolve C4 & C3: B and $\neg B \rightarrow \square$ (Empty Clause)

No further resolution is possible or necessary — the empty clause has already been reached.

Step 5: Final Conclusion

Because the Empty Clause ($\square$) was derived, the negated goal is inconsistent with the knowledge base. Hence the goal is PROVED:

$\therefore$ Priya can borrow books. (Goal successfully proved by resolution refutation.)

Question 4 — Placement Eligibility

Knowledge Base

- If a student clears the aptitude test, then the student is eligible for placement.
- Arun cleared the aptitude test.

Goal: Prove that Arun is eligible for placement, using the Resolution Algorithm.

Step 1: Propositional Logic Representation

- A = Arun clears the aptitude test
- P = Arun is eligible for placement

Premise 1: $A \rightarrow P$

Premise 2: A

Goal: P

Step 2: CNF Clauses

$A \rightarrow P: \equiv \neg A \vee P$

C1: $\neg A \vee P$

C2: A

Step 3: Negate the Goal

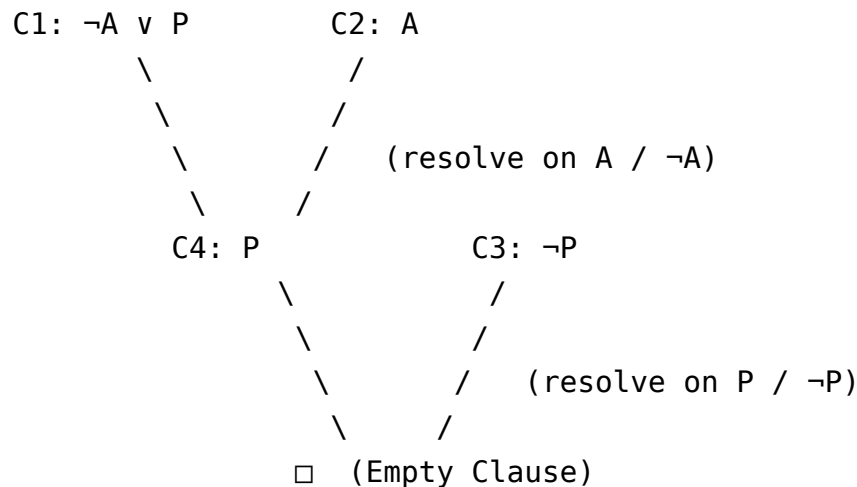
C3 ($\neg$ Goal): $\neg P$

Step 4: Apply the Resolution Algorithm Step by Step

Resolve C2 & C1: A and $\neg A \vee P \rightarrow P$ (call this C4)

Resolve C4 & C3: P and $\neg P \rightarrow \square$ (Empty Clause)

Step 5: Resolution Tree



Step 6: Conclusion

The resolution tree terminates in the Empty Clause ($\square$), proving that $\neg P$ is inconsistent with the knowledge base. Hence the goal is PROVED:

$\therefore$ Arun is eligible for placement. (Goal successfully proved.)

Question 5 — Access Control System

Knowledge Base

- If a user enters the correct password, then the user is authenticated.
- If a user is authenticated, then the user is granted access.
- The user entered the correct password.

Goal: Prove that the user is granted access, using the Resolution Algorithm.

Step 1: Propositional Logic Representation

- C = User enters the correct password
- A = User is authenticated
- G = User is granted access

Premise 1: $C \rightarrow A$

Premise 2: $A \rightarrow G$

Premise 3: C

Goal: G

Step 2: Convert All Implications to CNF

$C \rightarrow A: \equiv \neg C \vee A$

$A \rightarrow G: \equiv \neg A \vee G$

CNF clause set:

C1: $\neg C \vee A$

C2: $\neg A \vee G$

C3: C

Step 3: Negate the Goal

C4 ($\neg$ Goal): $\neg G$

Step 4: Apply Resolution to Derive the Empty Clause

Resolution proceeds in two chained steps because the chain password $\rightarrow$ authenticated $\rightarrow$ access has two implications:

Resolve C3 & C1: C and $\neg C \vee A \rightarrow A$ (call this C5; C / $\neg C$ cancel)

Resolve C5 & C2: A and $\neg A \vee G \rightarrow G$ (call this C6; A / $\neg A$ cancel)

Resolve C6 & C4: G and $\neg G \rightarrow \square$ (Empty Clause)

Step 5: Explanation of Each Resolution Step

- Step 1 uses the fact that the password was entered correctly (C) together with the rule $C \rightarrow A$ to derive that the user is authenticated (A).
- Step 2 uses the newly derived A together with the rule $A \rightarrow G$ to derive that access is granted (G).
- Step 3 resolves the derived G against the negated goal $\neg G$, producing the Empty Clause and completing the refutation.

Step 6: Final Conclusion

Since the resolution chain terminates in the Empty Clause ($\square$), the negated goal $\neg G$ is contradictory with the knowledge base. Hence the goal is PROVED:

$\therefore$ The user is granted access. (Goal successfully proved by resolution refutation.)

Summary of All Five Resolution Proofs

Every scenario in this assessment follows the same refutation pattern used in Question 1 through Question 5: convert the knowledge base to CNF, negate the goal, and resolve complementary literals until the Empty Clause ($\square$) is derived. The table below consolidates all five proofs for quick reference.

Q.No	Scenario / Goal	Premises (CNF)	Resolution Chain	Conclusion
1	Rain $\rightarrow$ Wet ground Goal: W	$\neg R \vee W, R$	$R + \neg R \vee W \rightarrow W$ $W + \neg W \rightarrow \square$	Proved: ground is wet
2	Submission $\rightarrow$ Marks Goal: M	$\neg S \vee M, S$	$S + \neg S \vee M \rightarrow M$ $M + \neg M \rightarrow \square$	Proved: Rahul gets marks
3	Membership $\rightarrow$ Borrow Goal: B	$\neg L \vee B, L$	$L + \neg L \vee B \rightarrow B$ $B + \neg B \rightarrow \square$	Proved: Priya can borrow
4	Aptitude $\rightarrow$ Placement Goal: P	$\neg A \vee P, A$	$A + \neg A \vee P \rightarrow P$ $P + \neg P \rightarrow \square$	Proved: Arun is eligible
5	Password $\rightarrow$ Auth $\rightarrow$ Access Goal: G	$\neg C \vee A, \neg A \vee G, C$	$C + \neg C \vee A \rightarrow A$ $A + \neg A \vee G \rightarrow G$ $G + \neg G \rightarrow \square$	Proved: access granted

General Pattern Observed

- Each knowledge base contains one or more implications of the form $X \rightarrow Y$, which convert to CNF as $\neg X \vee Y$.
- Each knowledge base also asserts the antecedent as a known fact (a unit clause).
- Resolving the unit fact against the implication clause strips the negated literal and yields the consequent as a new unit clause — this is effectively Modus Ponens performed through resolution.
- Chained implications (as in Question 5) require repeating this step once per link in the chain before the final contradiction with the negated goal is reached.
- In every case, reaching the Empty Clause ($\square$) confirms the knowledge base is unsatisfiable together with the negated goal — which, by the soundness of resolution refutation, proves the original goal follows logically from the knowledge base.

Overall Conclusion

All five goals — Q1: the ground is wet, Q2: Rahul receives internal marks, Q3: Priya can borrow books, Q4: Arun is eligible for placement, and Q5: the user is granted access — were each successfully proved true using the Resolution Algorithm, with every derivation terminating correctly in the Empty Clause ($\square$).